# Lee Wan Wei

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Technical Skills: Python, PyTorch, TensorFlow, scikit-learn, YOLOv8, OpenCV, RAG, LLM, FastAPI, Flask, Ruby on Rails, React, Next.js, Tailwind CSS

**Singapore University of Technology and Design**

- Bachelor of Science (Design and Artificial Intelligence) **Sep 22 to Present**

## WORK EXPERIENCES

**FPT Software** **Jul 25 – Sep 25**
**AI Engineer Intern**

- Addressed automotive QA inspection challenges by developing a defect detection system identify multiple defect types to support faster and consistent inspections by 70%
- Designed a **real-time decision-support dashboard** visualizing defects, severity scores, and trends for **internal QA stakeholders**
- Optimized inference pipelines to 45ms per frame to enable **low-latency, edge-compatible deployment**

**CyberG7 Technologies** **Aug 24 – Dec 24**
**AI Automation Engineer Intern**

- Developed automated data workflows to streamline business operations and reduce manual processing
- Integrated external APIs, LLM platforms, and automation tools to enable seamless data exchange across systems
- Evaluated and deployed tools for improved reliability, scalability, and operational efficiency

## PROJECTS

**AI Planting Visualizer**

- Designed and implemented an AI-driven web application that converts unstructured images and 2D spatial plans into structured landscape representations, reducing design time by 50%
- Built a multi-stage pipeline integrating image understanding, feature extraction, and RAG-based recommendation logic to generate plant suggestions under spatial, ecological, and design constraints
- Integrated open-source vision and NLP models to support explainable, data-driven planting decisions

**AI Text Training Simulator**

- Designed a **full-stack AI training simulator** enabling customer service teams to practice **simultaneous chat scenarios** grounded in real policy documents with about 2,000+ policy questions
- Engineered RAG-based retrieval pipelines to improve response accuracy and reduce hallucinations
- Implemented **rubric-based evaluation frameworks** to score responses on tone, accuracy, and comprehension, enabling structured performance feedback

**DBS Account Web Application Optimization**

- Developed a full-stack Ruby on Rails application using TDD/BDD practices to ensure reliable, iterative feature delivery
- Integrated OCR-based document extraction and OTP microservices to support secure identity verification and structured customer data workflows
- Improved data accuracy and onboarding efficiency through automated validation and verification pipelines

**Traffic Accident Hotspot Prediction**

- Implemented predictive models using geospatial, sensor, environmental, and time-series datasets to identify accident-prone road segments
- Engineered structured features and evaluated models using appropriate performance metrics to support robustness and interpretability
- Visualized hotspot risk scores and model outputs through structured graphs and dashboards for decision-making support

**SaveSmart – OCR Expense Categorization Application**

- Built a full-stack React + Flask application with an OCR pipeline to extract, clean, and structure receipt data
- Applied NLP-based classification models to automatically categorize expenses and generate analytics on spending patterns
- Improved expense tracking accuracy and reduced manual input through automated data processing

**Sentiment Analysis for Product Insights**

- Collected and processed large-scale text data from YouTube, Amazon, and Reddit using APIs and web scraping techniques
- Applied NLP sentiment analysis models to extract consumer opinions and actionable insights for product direction
- Translated unstructured feedback into structured signals to support data-driven product decisions